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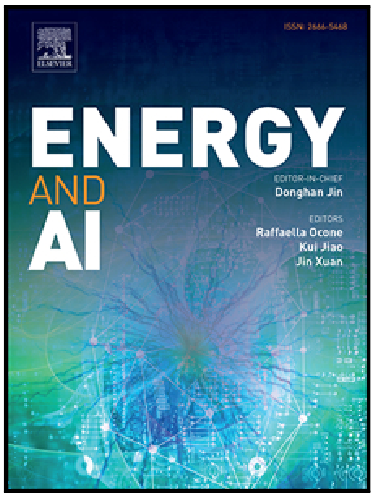

Can AI Be Energy Positive in Cement Manufacturing? Evaluating AI-Driven Efficiency Through a First-Order Framework and Case Study

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PII: S2666-5468(26)00115-1
DOI: https://doi.org/10.1016/j.egyai.2026.100789
Reference: EGYAI 100789

To appear in: *Energy and AI*

Received date: 17 December 2025
Revised date: 22 May 2026
Accepted date: 26 May 2026

Please cite this article as: M. Jibran S. Zuberi , Haitam Laarabi , Sarah Smith , Arman Shehabi , Can AI Be Energy Positive in Cement Manufacturing? Evaluating AI-Driven Efficiency Through a First-Order Framework and Case Study, *Energy and AI* (2026), doi: https://doi.org/10.1016/j.egyai.2026.100789

Highlights

- AI delivers net energy and emissions benefits in manufacturing.
- AI outperforms traditional controls in process optimization.
- Cement case study shows 3–9% net energy and CO_2 reductions.

Can AI Be Energy Positive in Cement Manufacturing? Evaluating AI-Driven Efficiency Through a First-Order Framework and Case Study

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Abstract:

The growing energy demands of artificial intelligence (AI) systems have raised increasing concern, yet their deployment in manufacturing may offer opportunities to improve efficiency, reduce costs, and enhance productivity beyond traditional control systems. This study presents a structured, first-order analytical framework for evaluating the net energy and environmental implications of AI deployment in industrial settings, demonstrated through a sector-specific case study of dry process cement manufacturing. The analysis examines AI use cases across multiple manufacturing sectors and cross-cutting utility systems, focusing on process optimization, energy efficiency potential, additional environmental and non-energy impacts, deployment archetypes, and supporting service providers. In the cement case study, where advanced AI services are emerging commercially, energy savings across selected unit operations are assessed alongside the electricity consumption of AI infrastructure, considering multiple server configurations and sensitivity analyses across deployment scenarios. Under explicitly defined, assumption-driven scenarios, the results suggest that the broader energy and economic benefits of AI deployment in manufacturing could outweigh the associated energy use. For the cement case study, these modeled scenarios correspond to a potential reduction in primary energy demand and associated CO_2 emissions of approximately 3–9%. These findings remain contingent on key assumptions regarding system boundaries, deployment configurations, and operational performance, and should therefore be interpreted as indicative rather than empirically validated outcomes. Overall, the proposed framework provides a methodological foundation for systematically evaluating AI's infrastructure energy requirements alongside its operational impacts, while the cement case study serves as an illustrative application to guide future empirical investigation and industrial decision-making.

Keywords: *Artificial intelligence, Deep Learning, Industry, Cement Manufacturing, Energy Footprint, Process Control.*

1. Introduction

Artificial intelligence (AI) is a broad and rapidly evolving field that encompasses a wide range of techniques and approaches. Historically, early AI systems were computer programs designed to replicate expert human decision-making for specific tasks. These systems relied on hard-coded rule sets and logical inference mechanisms to produce outputs that mimicked human reasoning [1]. Subsequent developments introduced heuristic methods, such as evolutionary algorithms, which can autonomously discover optimal solutions by maximizing performance criteria. More recently, the field has undergone a major shift with the rise of machine learning (ML), particularly deep learning (a subset of ML), and generative AI. These technologies are rapidly advancing while simultaneously gaining widespread adoption [2,3].

In parallel with these advancements, concerns have grown regarding the substantial energy consumption and associated emissions of AI systems. This issue has received increasing attention in both scientific literature and public discourse [4]. According to the International Energy Agency (IEA), global electricity demand from data centers is projected to double between 2024 and 2030, reaching approximately 945 TWh by 2030, with annual growth of around 15% [5]. Recent empirical studies have begun to quantify these energy and emissions impacts more rigorously. For example, Luccioni et al. (2023) [6] provide detailed estimates of energy consumption associated with AI inference, while Patterson et al. (2022) [7] analyze the emissions footprint of large-scale model training. These studies highlight significant variability in energy use depending on model architecture, deployment scale, hardware efficiency, and data center characteristics. However, the growing body of literature has primarily focused on the direct energy and emissions footprint of AI systems, with comparatively limited attention to their application-level net impacts in real-world sectors such as manufacturing. Focusing solely on the direct energy consumption risks overlooking potentially much larger indirect societal benefits, as emerging research suggests that, when deployed at scale, AI can significantly enhance macro-economic energy and resource efficiency, potentially leading to net reductions in overall energy demand and environmental impact, particularly in traditionally energy-intensive sectors like manufacturing [8].

Integrating AI into manufacturing represents a transformative step beyond conventional digitization and Information Technology (IT) integration. While digitization and IT systems provide foundational infrastructure, AI amplifies their value by generating actionable insights from large datasets. These insights enable more sophisticated process control, optimized preventive maintenance, and improved decision-making through advanced analytics [1]. Unlike traditional control systems, such as Programmable Logic Controllers (PLCs) and rule-based automation, AI systems provide adaptive, real-time decision-making that can respond to dynamic operational data, offering flexibility and responsiveness beyond pre-defined rules [9]. The deployment of advanced AI technologies in manufacturing holds significant promise for increasing efficiency, reducing costs, and enhancing productivity. Industry projections estimate that AI will contribute $10.5 billion in added revenue to the global manufacturing sector by 2033, an exponential increase from an estimated $0.33 billion in 2023 [10]. By leveraging deep learning and predictive analytics, AI systems can optimize industrial operations, detect anomalies, and enable real-time optimization across a range of manufacturing processes. Table 1 provides an overview of the key potential benefits of AI applications in manufacturing.

Despite these promising opportunities, a critical gap remains in understanding the net energy and environmental implications of AI deployment in manufacturing. Specifically, it is not well established whether AI applications in manufacturing are “energy positive”, i.e., whether the energy savings enabled by AI-driven process improvements exceed the total energy required to operate the AI system itself. In this study, an AI deployment is defined as energy positive if, on an annual basis and within clearly specified system boundaries, the energy savings at the process level exceed the combined energy demand of AI infrastructure, including model training, inference, and associated hardware. This definition provides a falsifiable condition for evaluating net impacts. However, given the variability in reported efficiency improvements (see Section 2.3) and uncertainty in AI energy use estimates, any such evaluation necessarily depends on explicit modeling assumptions and boundary definitions rather than universally generalizable empirical evidence.

Table 1 Key potential benefits of AI-applications in manufacturing (Sources: [1,11–14])

Benefits	Description
Improved efficiency and productivity	AI-driven automation streamlines production by handling repetitive tasks, minimizing errors, and optimizing workflows. Integrated systems enable hands-off manufacturing, from raw materials to final products, while real-time monitoring helps reduce energy inefficiencies.
Economic benefits	AI enhances automation, predictive analytics, and quality control to drive significant cost savings by reducing labor, maintenance, waste, and energy consumption, resulting in a leaner, more efficient production process.
Environmental benefits	AI supports sustainable manufacturing by optimizing resource use, reducing energy consumption and waste, and ultimately lowering environmental impacts through reduced emissions and improved waste management.
Increased safety	AI-powered collaborative robots (cobots) enhance workplace safety by handling hazardous tasks, while smart systems support precise and safe task execution alongside human workers.
Improved decision-making	AI enables real-time, data-driven decision-making, while digital twins simulate production scenarios to reduce risk and enhance planning.
Increased innovation and competitive advantage	AI accelerates innovation through faster prototyping, generative design, and digital twin simulations, helping manufacturers reduce time-to-market and develop advanced products to stay competitive.

Existing studies on the topic are limited in scope and often lack transparency in accounting for AI-related energy consumption. Many analyses focus on isolated case studies or report efficiency gains without systematically incorporating the energy use of AI infrastructure (see later sections for a detailed literature review). At the same time the literature on AI energy consumption, while increasingly robust, rarely connects system-level energy use to sector-specific operational benefits. This disconnect makes it difficult for policymakers, industry practitioners, and researchers to assess whether AI deployment leads to net energy and emissions reductions in practice. Consequently, the prevailing discourse tends to emphasize AI's energy footprint while underrepresenting its broader societal and economic benefits. Another key gap in the literature is the limited availability of data on the computational and energy requirements of AI services across different industrial applications. Addressing these gaps requires new insights into both the benefits and the energy trade-offs of AI deployment in manufacturing.

Building on the authors' previous work on data center energy consumption [15] and manufacturing sector assessments [16,17], this study aims to fill that gap by evaluating the net energy and environmental impacts of AI deployments in industrial settings through a structured framework and a sector-specific case study. More specifically, this research contributes to scientific literature in two main ways:

1. **Comprehensive review and qualitative assessment:** The study provides a thorough review of existing research, practical examples, and vendor claims, while offering an industry-wide qualitative assessment of AI applications in manufacturing. It examines how AI affects energy use, process efficiency, and environmental outcomes across various sectors and utility systems. It also identifies

key AI service providers, deployment archetypes, and operational parameters relevant to real-world use cases.

2. **Development of a net impact assessment framework:** The study introduces a simplified and transparent initial framework for evaluating net energy and emissions impacts within the plant boundaries, examined through a case study of dry process cement manufacturing. This analysis quantifies unit operation-level energy savings enabled by AI and compares them with the potential energy consumption by AI infrastructure. It models multiple server configurations and conducts sensitivity analysis to assess the effects of different deployment scenarios. The resulting approach allows broad applicability to other industrial contexts, offering a practical framework for preliminary impact assessments of AI services across various configurations and localities.

In summary, the key research gap lies in the absence of a quantifiable and transferable approach to assess whether AI-driven efficiency gains in industrial processes outweigh the energy requirements of AI systems. This study addresses this gap by providing a qualitative sector-wide assessment and demonstrating a simplified net-impact framework through a cement case study. It offers foundational insights and practical guidance for researchers and industrial practitioners, with implications that may inform policymakers in promoting holistic evaluation of AI adoption and supporting informed, responsible deployment in manufacturing.

The remainder of this article is organized as follows. Section 2 presents the materials and methods, beginning with a comprehensive literature review of AI technologies and algorithms used in manufacturing. It then narrows the scope to focus on the energy impacts of AI deployments in specific sectors and plant utility systems and subsequently outlines the net impact assessment framework through a case study on dry process cement manufacturing. Section 3 presents the qualitative assessment results, including operational benefits, use case archetypes, and AI model training times. This section also details the case study results, sensitivity analysis, and limitations, and provides recommendations for future research. Finally, Section 4 concludes the study by summarizing the key takeaways.

2. Materials and methods

This section outlines the scope and approach of the study. Literature and industry sources are systematically reviewed to identify AI technologies, use cases, and their reported impacts, forming the basis for a qualitative synthesis of benefits across sectors and utility systems. The section then narrows its focus to real-time operations in heavy industries, using cement as a case study, and describes a simple bottom-up model for quantifying plant-level net energy and emissions impacts resulting from AI deployment under transparent assumptions.

2.1. AI Technologies and Algorithms for Manufacturing Use Cases

Advanced AI technologies, including deep learning methods such as neural networks and generative AI, are set to transform every stage of the manufacturing process. These innovations enable smarter, more efficient, and highly adaptable plant operations across product design, process optimization, production, and quality control [1,9,10]. Specifically, deep learning involves algorithms that "learn" to identify patterns and make decisions by analyzing relevant data to solve specific tasks. These models are developed by gathering task-related datasets, selecting an appropriate learning method, and training the model accordingly. Deep learning algorithms can be broadly classified into three categories: supervised, unsupervised, and reinforcement learning [18].

Supervised learning trains models on labeled data, where inputs are paired with desired outputs, enabling the system to make predictions on new, unlabeled data. This approach is commonly used in image, speech, and object recognition. In contrast, unsupervised learning works with unlabeled data to discover patterns such as clustering similar items, reducing data complexity, or detecting anomalies without requiring explicit guidance. Reinforcement learning uses feedback from interactions with the environment to learn optimal actions through trial and error, making it especially effective for robotics and optimizing complex manufacturing processes [1,18].

Table 2 highlights key use cases of advanced AI in manufacturing. Building on this classification, the following discussion offers an overview of how deep learning and generative AI models can be applied in manufacturing to help stakeholders quickly grasp current research, relevant technologies and algorithms, and the overall direction of the field. It should be noted that the reviewed literature does not always provide quantified impacts for specific use cases based on modeling and/or measurements. However, where possible, quantified AI impacts using the algorithms discussed below are summarized for specific processes and their parameters in Section 3.1.

Table 2 General AI use cases in the manufacturing industry (Sources: [3,10,13,14,19])

Use case	Description
Process optimization and control	AI creates virtual replicas of processes and supply chains to simulate, monitor, and optimize performance and energy use in real time, reducing costs and environmental impact.
Collaborative robots (co-bots)	Co-bots work alongside humans, boosting productivity and safety in repetitive or strenuous tasks.
Quality control	AI, using computer vision and digital twins, detects product defects in real time with greater accuracy than human inspectors.
Predictive maintenance	AI forecasts equipment failures by analyzing sensor data, preventing downtime.
Custom manufacturing	AI-driven design adapts quickly to real-time consumer feedback for personalized production.
Supply chain management	AI enhances logistics by predicting demand, optimizing inventory, and improving efficiency.

Moreover, while design-phase and planning-stage AI applications, including generative design, custom manufacturing, and supply chain management, may have significant long-term impacts on material efficiency and lifecycle energy use, they fall outside the system boundary of the present analysis, which focuses on operational-phase energy impacts (see Section 2.2). Their inclusion in Section 2.1 is intended solely to provide a comprehensive view of the broader AI opportunity space in manufacturing and not to imply their consideration within the quantitative assessment.

2.1.1. Process Optimization and Control

Deep learning enhances manufacturing by improving equipment performance, product quality, and energy efficiency [20,21]. When integrated with automated control systems, it enables dynamic parameter adjustment, reduces manual intervention, and increases adaptability [19,22]. Deep Reinforcement Learning (DRL), including Q-learning, Deep Q-Networks (DQN), and Proximal Policy Optimization (PPO), has proven effective for optimizing real-time production parameters, ensuring efficiency and stability [23–25]. For instance, Ghaleb et al. [26] integrated reinforcement learning with Multiple Dispatching Rules (MDR) to improve real-time scheduling in failure-prone thermoplastic injection molding environments.

Recent advancements combining DRL with Graph Neural Networks (GNNs) support intelligent job scheduling and process planning. GNNs have been applied to model machining zones and optimize Computer Numerical Control (CNC) machining parameters [27,28]. Natural Language Processing (NLP) also contributes to manufacturing optimization by converting unstructured data into actionable insights. NLP techniques can extract information from maintenance reports or translate human language rules into machine-executable instructions, supporting intelligent scheduling and decision-making [29,30]. Despite these technological advances, challenges such as data quality, model generalization, and interpretability continue to hinder the widespread adoption of AI-based process control systems [19].

2.1.2. Collaborative Robots

Collaborative robots (or co-bots) work alongside humans to enhance productivity and safety by handling repetitive or physically demanding tasks, such as precise component placement in electronics manufacturing. By combining human dexterity with machine precision, co-bots bridge compatibility gaps [14]. Advances in AI, including reinforcement learning, computer vision, NLP, reasoning Large Language Models (LLMs) such as OpenAI's o3, multimodal AI such as Gemini 2.0, and systems like neuro-symbolic AI and GNNs, are rapidly increasing co-bot autonomy and adaptability [31,32]. For example, Gkournelos et al. [33] demonstrated a system called CollabAI, which integrates a fine-tuned GPT assistant into an augmented reality interface to assist with configuration and communication in automotive assembly. Similarly, Wang et al. [34] developed a vision-and-language co-bot navigation model that interprets verbal commands to generate autonomous navigation plans using the pathfinder algorithm, improving efficiency in tasks like tool retrieval. These studies highlight the growing role of LLMs in enabling more human-centric, adaptive capabilities in smart manufacturing environments.

2.1.3. Quality Control

Manufacturers aim to reduce production errors and defects [3], and AI models, particularly those using deep learning applied to image and audio data, have significantly improved defect detection and classification accuracy [35]. By automatically extracting features from large datasets, deep learning enhances both the efficiency and precision of quality control [36,37]. Recent literature has shown that Convolutional Neural Networks (CNNs) can match or even outperform traditional inspection methods in detecting surface defects. Zhang et al. [38] used CNN for surface defect detection in electronic components, while Essien and Giannetti [39] applied Long Short-Term Memory (LSTM) models for real-time fault prediction in manufacturing processes, reducing downtime. Stricker et al. [40] introduced an adaptive quality control method based on DRL, capable of adjusting strategies in response to real-time data. Together, these advances improve detection accuracy while increasing the efficiency, stability, and adaptability of production systems.

2.1.4. Predictive Maintenance

Predictive maintenance could leverage advanced AI to monitor equipment health, prevent failures, reduce downtime, and improve operational efficiency [41]. Traditional ML methods such as Support Vector Machines (SVM), Particle Filtering, Principal Component Analysis (PCA), Random Forests (RF), and Hidden Markov Models (HMM) have long supported predictive maintenance but often rely on manual feature selection and domain expertise, limiting scalability, response speed, and adaptability [19]. Deep learning models such as CNNs, Recurrent Neural Networks (RNNs), and LSTMs address these challenges, offering faster, more flexible solutions [42–44]. For example, Belmiloud et al. [45] combined Wavelet Packet Decomposition (WPD) with CNNs to extract features for predicting the remaining useful life of bearings.

2.1.5. Custom Manufacturing

AI is transforming manufacturing by enabling mass customization, allowing companies to tailor products to individual customer preferences without compromising production speed or efficiency. By integrating AI into the design process, manufacturers can rapidly adapt to real-time consumer feedback, enhancing both customer engagement and satisfaction. For example, in the textile and apparel industry, AI personalizes designs [14] using Generative Design (GD) technologies, which automatically produce multiple functional or aesthetic alternatives, speeding up the design cycle. However, traditional GD approaches often overlook manufacturability, which can limit their practical application [46].

To address this, Wang et al. (2025) [46] proposed a method that incorporates Deep Neural Networks (DNN), including CNNs and Generative Adversarial Networks (GANs), to embed design-for-manufacturing rules. This helps ensure that the generated designs are compatible with specific production processes. Similarly, Alfayoumi et al. [47] used Genetic Algorithms (GA) to optimize time and cost in mass-customized orders, grounding their method in real-world case studies and expert input. In another development, Zhang et al. [48] introduced a multi-agent system that combines edge computing with an enhanced contract net protocol, aiming to improve the efficiency and responsiveness of personalized manufacturing systems.

Generative AI design tools are increasingly applied in aerospace, automotive, and textiles, producing highly optimized parts and products that respect material and manufacturing constraints [14,46]. Platforms like NVIDIA Omniverse and ANSYS Twin Builder showcase the potential of AI-driven design [49], pointing toward a future of smarter and more agile production.

2.1.6. Supply Chain Management

Deep learning is also influencing supply chain management by improving material flows, waste management, and value-added activities [19]. AI models like LSTM enhance demand forecasting by utilizing time-series data to capture trends and seasonal patterns from historical sales data and can significantly improve forecast accuracy [50,51]. CNNs can match customer demand data with appropriate suppliers [52]. GNNs can reveal relationships among suppliers and facilitate company classification for better risk control and regulatory compliance [53]. Deep learning can also improve supplier selection by evaluating performance metrics such as price, quality, and delivery time [54].

Beyond forecasting and supplier selection, deep learning also optimizes production scheduling and inventory control. DRL has been applied to production planning to balance workloads and stabilize resource allocation [55], while simulations combined with deep learning can enhance batch sizing and scheduling under fluctuating demand and unreliable suppliers [56]. In inventory management, deep learning enables prediction of future inventory needs and supports automated replenishment, reducing the need for manual oversight [57,58]. Reinforcement learning can further optimize inventory strategies by dynamically adjusting safety stock levels and order quantities, addressing limitations of traditional models [59]. Despite its potential, deep learning application in supply chains remains limited due to scale, system complexity, and data diversity.

2.2. Scope of Work

The AI use cases in manufacturing, outlined in the previous section, are grouped in Figure 1 based on their focus on either design or operations. Design refers to the development, modeling, and testing of new or modified products and processes. In contrast, operations encompass the execution of manufacturing activities, involving machinery, sensors, control systems, human labor, infrastructure, and elements such as logistics and resource planning. Among these domains, product and process design have seen relatively greater industrial adoption of AI technologies [1], while AI-driven planning and real-time automation remain at earlier stages of implementation. In addition, product and process design activities are largely pre-commercial and are undertaken to enable efficient plant infrastructure prior to operations. As such, AI applications in design are generally not energy-related use cases; rather, they enable rapid innovation and scalable solutions, which are not the primary focus of this study.

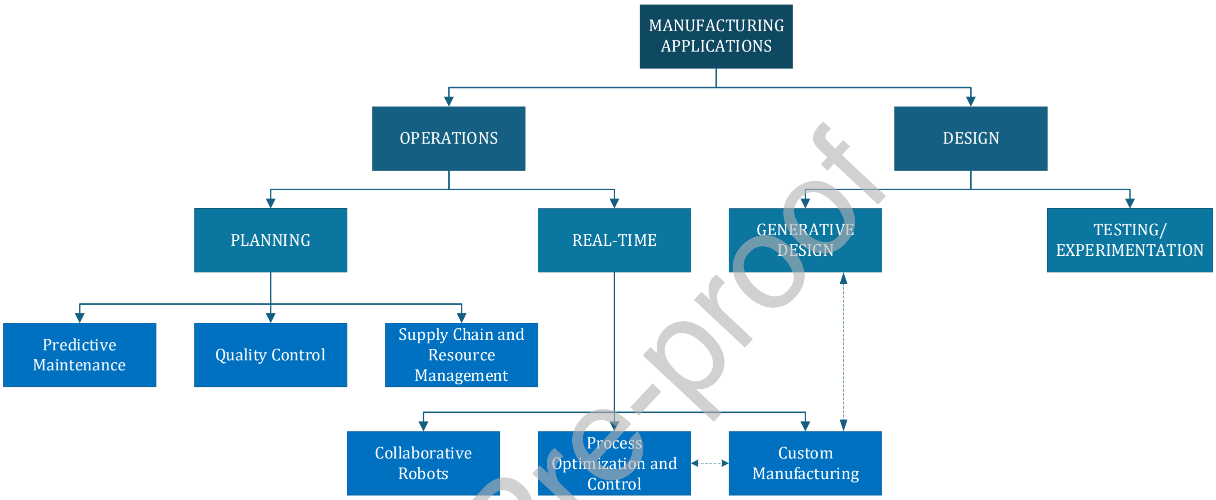

Figure 1 Classifying AI use cases in the manufacturing industry (partly adapted based on [1]).

AI applications have shown promise in improving efficiency, productivity, safety, and design quality, while also helping to reduce downtime. However, aside from energy use and carbon emissions, which are largely governed by thermodynamic constraints, most benefits are highly site-specific and difficult to generalize or quantify. For example, the impacts of predictive maintenance depend heavily on the level of system integration, plant configuration, and equipment vintage, which vary widely across industrial facilities even for the same process. As a result, the economic benefits associated with avoided breakdowns and reduced downtime from AI deployment are neither easily measurable nor readily generalizable. Moreover, there is currently no consensus in the literature on a suitable methodology to quantify the non-energy or co-benefits of emerging technologies and measures.

Accordingly, in line with the study's objective to evaluate the net energy and emissions impacts of AI deployment in manufacturing, specifically whether such deployments are energy positive or negative as discussed in Section 1, this study focuses on the energy and emissions impacts of AI-enabled real-time operations. However, where possible, qualitative assessments of non-energy benefits such as productivity improvements and material efficiency are also included.

Real-time operations span both light and heavy manufacturing. Light manufacturing sectors, such as automotive, textiles, and machining, typically involve physical transformations and have been early adopters of AI technologies. However, plant-level energy intensities in these sectors vary significantly and lack standardization, making the impacts of AI deployment inconsistent and difficult to compare across sites. Heavy manufacturing sectors, including chemicals, iron and steel, and cement, are characterized by complex chemical and physical transformations that pose substantial operational risks. These industries are also far more energy- and emissions-intensive than light manufacturing. Although baseline process intensities in heavy manufacturing are relatively well-documented, the adoption of AI technologies remains limited and underexplored.

To address the first objective in Section 1, this study presents a comprehensive review of existing research and practical examples, providing a sector-wide qualitative assessment of real-time AI applications in manufacturing. Key factors identified include process efficiency, environmental outcomes, AI service providers, deployment models, and operational parameters relevant to real-world use cases across different sectors and utility systems. It should be noted that the reviewed literature comprises a diverse mix of sources, including academic publications, technical reports, and vendor claims.

The level of confidence in reported efficiency improvements, environmental benefits, and other impacts varies across these sources. Commercial reports are generally grounded in empirical operational experience, whereas findings from academic studies may be inconsistent, subject to publication bias, or based on theoretical models that may not fully translate to real-world applications. Meanwhile, vendor claims may carry inherent bias due to their promotional nature. To systematically address these variations and uncertainties, the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework is used to assess and assign confidence levels to the reported impacts of each AI application reviewed. Further details on the GRADE framework can be found in Prasad (2024) [60]. Using this framework, impact certainty for commercially demonstrated and independently validated results can be rated as "High," while modeled impacts reported in academic literature are rated as "Moderate." If commercially reported impacts are not independently validated, their impact certainty is also rated as "Moderate." Vendor claims are assigned a "Low" impact certainty, and claims that aggregate impacts across multiple unit operations or processes are assigned a "Very Low" impact certainty.

To estimate the quantifiable impacts of AI in high-risk, high-reward industrial environments, the cement manufacturing sector has been selected as a case study. Cement production is notable as one of the few heavy manufacturing sectors where AI services have already been developed, trained, and commercialized, making it both a relevant and timely example, given its substantial energy demands and environmental footprint. Despite these advancements, there is still a notable lack of concrete data on the computational demands of AI applications within manufacturing settings. This data gap necessitates broad assumptions, which introduce a degree of uncertainty into any analysis. Specifically for cement manufacturing, little information is available in the literature regarding the computational infrastructure required to support AI systems. To address this, the study introduces what may be the first set of assumptions on AI infrastructure in this context, accompanied by sensitivity and uncertainty analyses to estimate the potential range of net energy impacts from real-time AI applications in operational environments; non-energy benefits are outside the scope of the case study.

2.3. Net Impacts of Real-Time AI Applications in the Manufacturing Industry

To develop a simplified initial framework for assessing the net energy and emissions impacts of AI applications at the sectoral level, this study uses cement manufacturing as a case study. The analysis begins by establishing process energy intensities for a representative plant configuration, disaggregated at the unit operation level. The reference case is a cement plant employing the dry process with a preheater and precalciner kiln system, currently the most common and state-of-the-art technology globally, including in the U.S. This plant is assumed to have a clinker production capacity of 1 million tonnes (Mt) per year, with a primary energy intensity of 1,210 GWh_{th} per tonne of clinker (4.4 GJ/t-cl) [16,61,62] and associated energy-related CO_2 emissions of approximately 0.4 Mt CO_2.

Figure 2 presents a simplified block flow diagram of the typical dry process cement manufacturing plant. It also illustrates the relative contributions of each unit operation to the total plant primary energy intensity, based on data from prior work partly conducted by the authors using literature sources [16,61,62]. All energy values are expressed in primary energy terms using a 2022 primary energy factor of 2.6 based on the U.S. EIA 2023 Annual Energy Outlook (AEO) [63]. It should be noted that this primary energy factor reflects the U.S. national average and may vary significantly depending on the regional mix of electricity generation technologies, however, geospatial and regional analyses are currently outside the scope of this work. Furthermore, it is assumed that on-site AI deployment (via micro data centers; see the following discussion) will source electricity from the same grid as the plant's other unit operations, therefore, the same primary energy factor is applied.

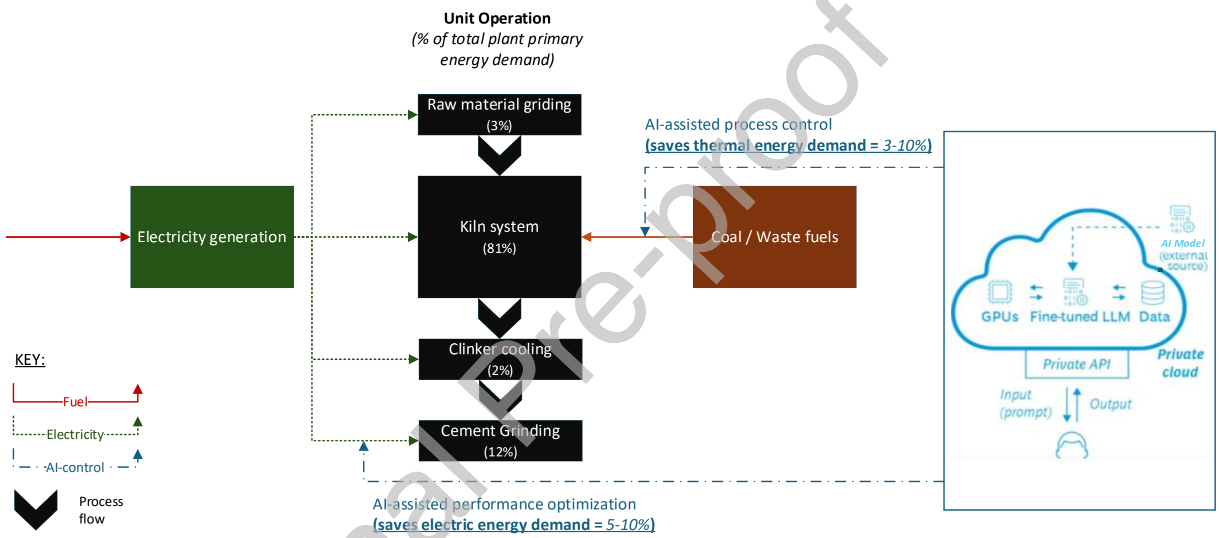

Figure 2 Simplified block flow diagram of AI-assisted cement manufacturing (partly adapted from [9,16]).

Figure 2 also highlights unit operations for which AI-assisted process improvements have been reported in the literature. The unit operation-level energy savings enabled by AI *'ES_P'* are derived from a comprehensive review of AI applications in the cement sector (see Section 3.1). While some literature claims savings exceeding 10% for kiln systems, the values presented in Figure 2 and used in this analysis (3% gross savings representing a conservative case and up to 10% representing an optimistic scenario) are grounded in real-world commercial experience rather than vendor projections. For example, McKinsey analyzed a cement kiln and mill in North America operating in autonomous mode using AI and reported energy efficiency improvements of up to 10% [64]. Similarly, Heidelberg Materials implemented AI-based advanced process control at a cement plant in Europe, achieving a 4.1% reduction in fuel costs and reducing specific heat consumption by 2–3%. Together, these examples present a realistic range of energy savings that can be assessed using upper and lower bounds. Accordingly, *'ES_P'* reflects reported savings from a limited number of commercial AI deployments and should not be interpreted as guaranteed or universally persistent across facilities. Despite this uncertainty, the full range of reported savings potential shown in Figure 2 is used to define a plausible range of outcomes. These savings are then compared with the energy demand of the AI infrastructure 'ED_{AI}' to estimate the net energy impact of AI-enabled services using Equations 1 and 2.

$$ES_N = ES_P - ED_{AI} \tag{1}$$

Where,
ES_N = Net energy impact of AI services
ES_P = Total energy savings from AI applications
ED_{AI} = Energy demand by AI technologies

$$ES_P = \sum ES_U \tag{2}$$

Where,
ES_U = Energy savings by unit operation due to AI applications

The increased complexity of the technological foundation required to effectively deploy AI in manufacturing has led to distinct operating models. The simplest approach relies on fully provider-based solutions, offering straightforward integration. Alternatively, manufacturers may fine-tune open-source models or develop custom AI solutions in-house on their own infrastructure. While these latter options enable greater customization and enhance data security, they require significant investments in computing hardware, such as Graphics Processing Units (GPUs), and demand specialized expertise to design, operate, and maintain the systems effectively [9].

Manufacturing processes are faster than ever, generating unprecedented volumes of data from sensors, inspection images, and process metrics. Critical decisions made in the moment between data collection and action directly impact product quality, equipment lifespan, and profitability. Traditional data handling methods often cannot keep pace, and while cloud systems offer substantial processing power, their physical distance from the production facility introduces delays that are incompatible with time-sensitive operations. On high-speed production lines, even seconds of delay can lead to defective products, equipment damage, or costly coordination ripple effects [65].

Edge computing addresses this challenge by processing data locally, bringing AI-enabled computing power directly to the production facility. Industrial-grade edge systems enable real-time decision-making in microseconds, supporting process optimization, predictive maintenance, and quality control [65]. Given the energy-intensive and complex nature of sectors such as cement, industrial AI workloads are expected to be substantial, making edge computing essential. Due to limited publicly available data on AI computational demands in manufacturing settings, it is assumed that cement plants prioritize data security and operational control by deploying on-site AI solutions, effectively adding the AI energy demand to the facility's overall intensity and allowing direct comparison within the plant boundaries. AI models for cement plants are assumed to be pre-trained and fine-tuned locally on multiple servers via private cloud solutions, as illustrated in Figure 2. While alternative vendor deployment models exist, these are discussed qualitatively in Section 3.1.1.

Hybrid architectures are also emerging, in which edge systems handle time-sensitive inference while cloud platforms manage aggregation, long-term analytics, and large-scale model training. This approach balances the immediacy of edge computing with the scalability of the cloud, especially as AI integrates with Industrial Internet of Things (IoT) to enhance operational visibility [65,66]. In the absence of concrete data on server requirements for AI in cement plants, a sensitivity analysis considers 1-5 servers per plant, each using 2–8 GPUs or non-accelerated AI, with the upper range included to account for variability and uncertainty. In addition, to account for the energy use during the AI model's pre-deployment training period, it is assumed that an 8-GPU server runs continuously for 1–3 months (empirical range; see Section 3.1.3 and Table 5 for details), with a PUE of 2 (reflecting a heavy computational load that requires relatively more cooling than inference), based on the discussion in Section 3.1.3.

Figure 3 shows the average power draw of various server types from 2018 to 2023, with projections through 2028. The average power draw for each server type was estimated using the servers' peak power ratings, typical utilization across various workloads, and in-use power measurements that reflect actual consumption relative to peak under different conditions, refer to Shehabi et al. [15] for a detailed description of the method and assumptions underlying Figure 3. For this analysis, 2023 average values are used as the baseline, assuming continuous operation (8,760 hours annually). In addition, terabytes (TB) of plant process data must be managed and stored [67]. The server systems shown in Figure 3 include both storage and networking equipment. For example, the NVIDIA DGX H100/H200 systems, which are based on 8 GPUs, provide approximately 30.72 TB of high-performance Non-Volatile Memory Express (NVMe) storage dedicated to application data and caching [68]. This capacity is assumed to be sufficient for inferencing and periodic training. However, additional storage may be required for retaining historical plant data, which could be managed using external storage drives or accessed via the cloud. It should be noted that any external storage (if required, as this remains uncertain) would introduce additional power consumption that is not currently accounted for.

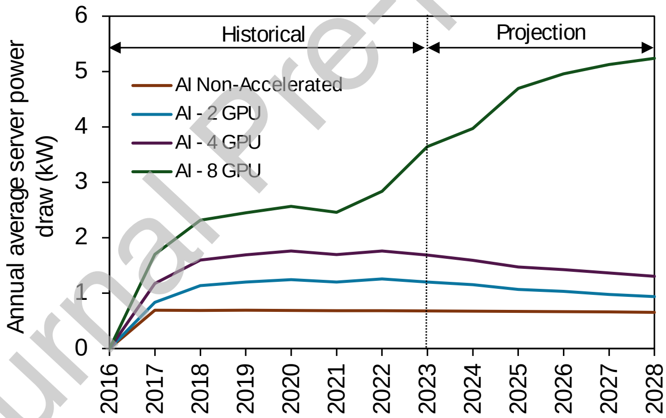

Figure 3 Aggregate average power draw of various server types by year (recreated from [15]).

Furthermore, while server cooling will require electricity, it can be assumed that localized cooling, such as air conditioning in plant control rooms, will suffice given that the servers are not housed in large scale data centers. However, for the purposes of this case study, a micro data center (MCD) approach is assumed. An MCD is defined as a compact, self-contained unit that integrates essential data center components, such as servers, storage, networking equipment, and cooling, into a single server rack. Conservative Power Usage Effectiveness (PUE) values of 1.5 to 2 are assumed for the MCD to represent lower and upper bound estimates, respectively (with the lower PUE corresponding to maximum net energy savings, and vice versa). In practice, more efficient MCD designs may achieve lower PUE values, which would further improve the net energy impact relative to the current assumptions.

Since reported relative energy savings from AI applications in cement kilns and grinding vary widely, and server energy demand is based on scenarios ranging from 1–5 servers per plant (either 2–8 GPUs or non-accelerated AI servers with PUEs of 1.5- 2, typical for small server rooms [69] representative of industrial plant-scale MCD, and a pre-deployment training period of 1-3 months based on Carbon Re's AI deployment at Heidelberg Materials' cement plant in Europe [70,71]), an uncertainty analysis is necessary. To address this, we perform Monte Carlo simulations using triangular probability distributions to represent uncertainty in plant energy savings and AI energy demand, reflecting bounded operating ranges and specified most-likely values rather than assuming unbounded normal behavior. A total of 10,000 simulation runs were conducted to generate a distribution of potential net energy savings outcomes under realistic operating conditions. Triangular distributions were selected because the underlying variables are physically constrained and are more appropriately characterized by minimum, maximum, and most-likely values. To provide a conservative assessment, the most-likely plant energy savings were assumed to be equal to the lower bound (35.83 GWh_{th} per year), reflecting the deliberately cautious assumption that lower savings are more probable. In contrast, the most-likely AI energy demand was set at 0.17 GWh per year, representing one 8-GPU server operating at a PUE of 2 for inferencing and periodic training, based on expert judgment regarding plausible operational AI workloads in manufacturing.

Furthermore, based on the aforementioned assumptions and methodology, the net energy savings from AI applications in the cement sector can be estimated. These savings are then translated into net CO_2 abatement 'CA_N' using Equation 3 and a weighted average fuel emission factor *'EF'* of 0.34 t-CO_2/MWh_{th} for kilns and 0.31 t-CO_2/MWh_{th} for the electricity grid, consistent with the 2022 cement sector fuel mix data from MECS [72] and the 2022 grid fuel mix from AEO [63], respectively. Cement plants are assumed to purchase electricity from thermal power sources, excluding nuclear and renewables, therefore, the grid mix reflects a conservative scenario. However, an alternative perspective is also briefly provided by considering the overall grid mix, including non-fossil energy sources such as nuclear power. It should be noted that cement manufacturing also generates significant process emissions, approximately 0.51 t-CO_2 per tonne of clinker [73], which account for roughly 50–60% of total plant emissions [74]. These emissions arise primarily from calcination, a chemical process in which limestone ($CaCO_3$) is heated and decomposed into lime (CaO) and CO_2. However, process emissions are inherently linked to process chemistry and production volume. Since the AI-enabled efficiency improvements in this study are assessed on a per-tonne-of-cement basis, process emissions remain unchanged with AI deployment. Consequently, the net CO_2 abatement estimated here is driven solely by reductions in fuel consumption and associated combustion-related emissions.

$$CA_N = ES_N \times EF \tag{3}$$

Where,
CA_N = Net CO_2 impact (abatement) of AI services
EF = Weighted average fuel emission factor

3. Results

3.1. Qualitative Assessment of AI-driven Real-Time Applications in Manufacturing

3.1.1. Potential Applications in Specific Manufacturing Sectors

Although advanced AI technologies are still maturing, the manufacturing industry and service providers are already investing in selected use cases to leverage AI's ability to process large datasets and deliver actionable intelligence, particularly in real-time operations and process optimization. Table 3 highlights several early-stage applications across specific manufacturing sectors where such investments and implementations are currently taking place. These examples illustrate how different sectors are beginning to integrate AI into their operations, with the potential for broader adoption across production and supply chain management. While Table 3 is not exhaustive, it reflects some of the most prominent applications documented in academic literature and corporate materials. These cases provide a qualitative assessment of AI's impact on enhancing real-time operations through improvements in energy efficiency, CO_2 abatement, and other non-energy-related benefits.

The deployment of advanced AI technologies across industrial processes indicates a consistent pattern of both energy and non-energy benefits, despite sectoral differences. The review further indicates that manufacturing plants are adopting AI to improve process efficiency, stabilize operations, reduce energy consumption, and enable real-time decision-making through advanced analytics. Across diverse industrial contexts, a consistent set of applications has emerged, including the optimization of dynamic process parameters, predictive quality control, anomaly detection, and equipment fault identification. AI is showing particular effectiveness in optimizing complex, multivariate systems that require constant adjustments to maintain operational stability. In several industrial processes, AI technologies such as neural networks are being used to dynamically adjust key variables like flow rates, feed ratios, temperature, and pressure. These AI-enabled control systems are suggested to improve energy efficiency while also enhancing productivity. Furthermore, AI models and predictive analytics are shown to reduce unplanned downtime and operational costs. Techniques such as Long Short-Term Memory (LSTM) networks, Random Forests (RF), and Gradient Boosting Regression (GBR) may enable accurate forecasting of system behavior and early fault detection. These capabilities could contribute to enhanced system reliability, improved safety, and optimized use of resources such as fuel, catalysts, and raw materials.

Despite these promising outcomes, the maturity of AI applications varies significantly by context. While some implementations show measurable improvements in energy use, productivity, and cost efficiency, others remain in experimental or early adoption phases. In some cases, evidence of scalability and generalizability across different settings remains inconclusive, indicating a need for further empirical validation. For example, service providers such as ABB, ThyssenKrupp, CarbonRe, and Basetwo have reported AI solutions for major cement plant operations, including kiln systems and cement grinding [70,75–77]. However, the reported impacts on energy efficiency vary widely (from 3% to 10% of reference energy demand, as commercially reported), possibly due to differences in plant configurations, levels of heat integration, and the effectiveness of the AI models themselves. Additional factors may also play a role; for example, cement facilities might prefer AI models operated through privately controlled on-site systems due to confidentiality and security considerations. In contrast, CarbonRe's solution requires no systems integration, hardware investments, or new equipment [70]. It uses a cement plant's digital twin to evaluate performance and provides cloud-based recommendations for optimization parameters that can be implemented through existing control systems. These diverse application models and archetypes, along with the lack of a consistent framework for assessing net impacts, make it difficult to generalize the results.

Similarly, the positive impacts of AI applications in the chemical sector, especially those noted in Table 3, largely pertain to petrochemicals. However, the chemical sector in general, and the petrochemical industry in particular, is highly diverse in terms of unit processes and thermodynamic requirements, making it difficult to generalize reported impacts across the sector. That said, based on similar optimization challenges, AI applications not yet reported in some sub-sectors could plausibly be applied in others. For example, AI tools used to optimize fuel mixes for kiln stability in cement manufacturing could be adapted for use in iron and steel, lime, or steam cracking processes, where variable fuels are commonly employed for process heat and availability is constrained by feedstock and logistical factors.

Among the non-energy benefits, visual detection of product anomalies is another transferable application. Powered by GPUs and CNNs, AI systems can scan automotive parts at 2,000 frames per second with 99.98% accuracy [78]. For example, BMW's pilot in Leipzig integrated vision AI into its bumper production line and reduced scrap rates by 34%, identifying defects as subtle as 0.1 mm that are often missed by human inspectors under factory lighting [78]. Similar AI-based anomaly detection algorithms could be applied across the light manufacturing sector beyond automotive for quality assurance in any assembly line involving physical transformation. Injection molding, a widely used process, is another area with potential for AI-driven optimization across multiple sectors wherever applicable.

Table 3. Reported early-stage applications of advanced AI in specific manufacturing sectors.

Use case	**Industrial sector**	**Parameters optimized through AI technologies**	**Service providers**	**AI system**	**Reported relative energy savings and other impacts**		**Sources**	**Potential sectors relevant to the use case**
					Potential impact	**Impact certainty**		
AI-assisted process control, stabilization, and optimization.	Cement	***Kiln system:*** Rates of alternative fuel substitution must be optimized to ensure consistent combustion while avoiding instability caused by fluctuations in the fuel's calorific value. Specific parameters subject to optimization include temperature control, rates of alternative fuel substitution, kiln speed, feed rate, and airflow. ***Cement grinding:*** Raw material proportions, speed in which material is fed (feed rate), how fast the separator is moving, and fan speeds for the main ventilator.	ABB, ThyssenKrupp, CarbonRe Imubit Basetwo	Linear and Non-Linear Model Predictive Control, Fuzzy Logic and Artificial Neural Networks (ANN)	Achieves approx. 3–20% reduction in overall energy demand and boosts overall productivity by 3–8%. *(Up to 10% energy reduction has been commercially realized in a few cement plants, though not independently validated, while the projected 20% reduction remains a vendor claim)*	Moderate	[64,70,71,75–77,79–82]	Iron and steel (coal, coke, blast furnace gas, coke oven gas, natural gas, etc.), steam cracking (natural gas, tail gas), and lime manufacturing sectors are relevant for fuel mix optimization.

AI-assisted optimization of process parameters, variable product yields, and real-time decision-making, considering equipment constraints	Chemicals and Refineries	Deep learning and predictive analytics to optimize process parameters such as reactor outlet temperature (ROT), pressure, reaction rates, feed composition, feed rate; detect anomalies; and facilitate real-time decision-making.	Imubit	Deep Learning Process Control (DLPC), Long Short-Term Memory (LSTM), Recurrent Neural Networks (RNN)	Enhances process efficiency by 12–18%, lowers operating costs by 16–25% and cuts downtime by 22–30%. AI-driven automation accelerates decision-making by up to 50%, improves catalyst performance by 30%, and boosts cybersecurity resilience by 28%. *(peer-reviewed literature-reported)*	Very Low	[83–85]	All chemical and petroleum refining processes.
AI-driven optimization of production parameters leading to efficiency gains	Biodiesel and Glycerol	Optimizing oil-to-alcohol ratio, process temperature, time to complete the process, and type and amount of catalyst.	N/A	ANN	15-20% of overall energy demand. *(peer-reviewed literature-reported)*	Moderate	[86,87]	Inconclusive
AI-assisted quality control and process optimization	Pulp and Paper	Predict quality parameters like folding endurance, bursting strength, and smoothness. Detects surface defects like tears more accurately than human eyes and adjusts machine settings to optimize process parameters like stirring speed, water use, air pressure, and exhaust temperature.	EasyODM	Random Forest (RF) and Gradient Boosting Regression (GBR) models; the application of ANN is currently being researched.	Achieves approximately 17% reduction in overall energy demand. *(vendor-reported)*	Low	[88,89]	Inconclusive
AI-driven optimization of the BOF plant for	Iron and Steel	Optimizing BOF gas flow rate, air infiltration during blowing, as well as oxygen and temperature levels.	ArcelorMittal Aviles	Bio-inspired Ant Colony Optimization (ACO), along	BOF gas recovery rate increased by 49%, and its caloric value increased by	Moderate	[90–92]	Inconclusive

optimal steam and gas generation and utilization				with other techniques like CNN and LSTM.	14%, providing more fuel energy to offset natural gas use. *(commercially realized; not independently validated)*			
AI-driven visual defect detection	Automotive	Defect detection, including sub-millimeter weld porosity, paint runs, hairline cracks, and other flaws, using high-resolution cameras.	Matroid	Automated visual inspection with CNNs.	Reduce scrap rates by up to 34% *(commercially realized; not independently validated)*	Moderate	[78,93]	All light manufacturing that requires product visual inspection
AI-driven injection molding to control tensile strength and fatigue resistance	Polymers / Plastics	Optimization through dynamic adjustment of shot size based on real-time resin density fluctuations, stabilization of melt viscosity, and control of barrel temperatures, screw speed, filler content, and other parameters.	Siemens, ABB	ANN	Energy use decreased by 9% and reduced cycle times by 15%. *(commercially realized; not independently validated)*	Moderate	[78,94]	All sectors that involve injection molding

Note: The use cases and corresponding impacts do not represent a single AI product. Instead, each row presents a summary of relevant parameters and impacts as reported by various literature sources and service providers. Additionally, energy savings and other impacts for the cement, steel, automotive, and polymer sectors are reported as commercially realized, whereas for other sectors it remains inconclusive whether the reported impacts are modeled, realized, or inferred.

3.1.2. Potential Applications in Specific Manufacturing Utility Systems

Beyond specific industrial sectors, the growing adoption of AI in plant utility systems across the manufacturing industry highlights the increasing importance of intelligent control and optimization in areas that have traditionally been passive and manually managed. Systems such as cooling, heating, ventilation, and lighting are evolving from fixed rule-based management to adaptive, data-driven operations that significantly enhance energy efficiency and overall performance.

Table 4 presents early reported applications of advanced AI in specific utility systems. In these cases, AI is primarily used to enable predictive control, adaptive operation, and precise optimization of energy demand. By leveraging continuous data from occupancy sensors, weather conditions, equipment performance, and user preferences, AI technologies can dynamically adjust plant utility operations to match current loads while minimizing unnecessary energy consumption. Techniques such as neural networks, genetic algorithms (GAs), and fuzzy control methods are commonly employed to model and manage system behavior with a high degree of accuracy.

The advantages of AI integration in utility systems are particularly reflected in energy savings. In the manufacturing sector, approximately 70% of final energy demand is used for process heat [95], with nearly 50% of this heat delivered as steam produced by boiler systems [96]. Cooling or refrigeration is another essential utility widely used across industries, particularly in the food and beverage and pharmaceutical manufacturing sectors. In both cooling and boiler systems, AI has been shown to improve load distribution, ensure operational redundancy, and manage capacity more efficiently. These enhancements have resulted in reported energy savings of approximately 10–15% compared to the reference energy demand of utility systems. Specifically for steam generation, AI methods that combine ANN, GA, and Computational Fluid Dynamics (CFD) models could improve thermal efficiency and reduced NOx emissions [91,97,98].

In Heating, Ventilation, and Air Conditioning (HVAC) systems, AI-driven solutions that respond to real-time data on occupancy, weather conditions, and operational requirements can deliver substantial efficiency gains, with reported reductions in utility energy demand ranging from 10 to 40%. However, these reported values are drawn from a mix of applications across commercial, institutional, and industrial settings, as manufacturing-specific evidence remains relatively limited. This wide range depends on several factors, including system configuration, climate zone, and facility vintage. In addition, the achievable savings are influenced by the specific AI approaches used for learning, optimization, and control, as demonstrated by Lee et al. [99]. Similarly, intelligent lighting systems can deliver significant energy savings by adapting to daylight availability, user activity, and occupancy patterns, with reported reductions in lighting energy use ranging from 11 to 50%, although much of the available evidence also originates from non-manufacturing environments. While HVAC and lighting systems typically represent a smaller share of total facility energy demand compared with process energy use, they remain relevant cross-cutting utility systems that influence indoor environmental quality, workplace comfort, and employee safety.

Moreover, early applications of AI in utility systems further suggest how automation can extend beyond core manufacturing processes to enhance the efficiency of entire facilities. For instance, Schneider Electric's AI-based optimization platform [100] offers an integrated view of various energy utility components, combining intelligent recommendations with cost tracking to support more informed operational decision-making. Qualitative findings across different sectors and utility systems emphasize AI's growing role in transforming industrial operations. Its ability to analyze complex data, adapt to changing conditions, and uncover new optimization strategies positions AI as a key driver of smarter, more sustainable manufacturing environments. To fully realize AI's potential, ongoing investment in research, broader system integration, and effective knowledge sharing across industries will be essential.

Table 4. Reported early-stage applications of AI in specific manufacturing utility systems.

Use case	Utility system	Parameters optimized through AI technologies	Service providers	AI system	Reported relative energy savings and other impacts		Sources
					Potential impacts	**Impact certainty**	
AI-driven optimization of energy use in plant cooling and steam utilities.	Cooling and boiler systems	Predict plant peak load, chiller load ratio, heat exchanger demand, temperature corrected capacity in use, etc. Schneider Electric's EcoStruxure Industrial Advisor – Predictive Energy platform optimizes energy usage, temperature, flow, and pressure, ensuring efficient operation of manufacturing utilities.	Schneider Electric	ANN, Genetic Algorithm (GA), LSTM	10-15% of energy use by relevant utility systems *(vendor-reported)*	Very low	[91,100–102]
AI-assisted HVAC control	HVAC	Dynamically adjusts HVAC operations based on real-time data such as occupancy, weather conditions, and energy costs. For instance, it reduces heating or cooling in unoccupied spaces, minimizing waste.	N/A	Future prediction and pre-adjustment with Multi-Layer Perception (MLP) / Feedforward Neural Networks (FNN) for model learning	10-40% of the HVAC energy demand *(peer-reviewed literature-reported)*	Moderate	[99,103]
AI-assisted personal interactive lighting control	Lighting	Adjust lighting based on occupancy, daylight availability, or user preferences. For example, lights can dim or switch off in unoccupied areas to save energy.	N/A	Convolutional Neural Network (CNN), Adaptive Neuro Fuzzy Interference System (ANFIS), and Fuzzy Control	11-50% of the lighting energy demand *(peer-reviewed literature-reported)*	Moderate	[99]

<u>Note:</u> The use cases and corresponding impacts do not represent a single AI product. Instead, each row presents a summary of relevant parameters and impacts as reported by various literature sources and service providers. Additionally, for all utility systems, it remains inconclusive whether the reported energy savings are modeled, realized, or inferred.

3.1.3. AI Model Training Times and Inference

In reinforcement learning, training refers to the process by which an AI model improves its decision-making through repeated interactions with a simulated or real-world environment. Through these interactions, the model learns to select actions based on context, gradually refining its neural network. The learning process is guided by a reward function, which is designed to promote desired outcomes. For example, in optimizing cement plant operations, a reward function might include carbon cost considerations, encouraging the model to favor low-emission decisions that still meet production targets. It may also account for safety constraints, such as temperature or pressure thresholds. Despite its sophistication, reinforcement learning is inherently brute-force, with models exploring a vast array of scenarios by experimenting with different actions to understand their consequences. Over time, this trial-and-error approach enables the AI to uncover strategies aligned with long-term goals. In cement manufacturing, for instance, this could involve dynamically adjusting fuel distribution between the pre-calciner and kiln in response to fluctuations in the raw feed's burnability. Such insights are made possible by advances in computing power and high-speed processors like GPUs, which enable AI to simulate years of plant operation in just days or weeks.

Training AI models for manufacturing applications demands both substantial data movement (e.g., memory transfers) and intensive computation (e.g., high-dimensional matrix operations) [1]. Depending on model complexity, training can span from several hours to months and may consume significant energy. While training has historically been the dominant contributor to AI-related energy use, recent studies show that energy consumption varies widely depending on model scale, architecture, and deployment context [6,7]. However, regardless of scale, when powered by non-renewable sources, AI-related energy use can result in considerable emissions. For example, one study found that training a DNN-based NLP model could generate emissions comparable to nearly 60% of those produced by an average car over its lifetime [1]. AI models used in industrial and manufacturing settings are typically several orders of magnitude smaller and are often trained on domain-specific datasets, resulting in substantially lower computational and energy requirements [104]. While inference is generally less energy-intensive than training, the energy required per inference step in DNNs has increased in some cases, from approximately 0.1 J to 20 J between 2012 and 2022 [105]. These trends underscore the environmental implications of advanced AI models when deployed without consideration of renewable energy sources or their net energy impact in manufacturing contexts.

Table 5 presents training time estimates for AI models across various manufacturing use cases, ranging from several days to months depending on the model type and application. In general, training duration is influenced by use case complexity, the number of variables involved, and the nature of their interactions. While research continues to explore process-specific challenges, industry literature indicates that AI service providers are developing generalized models for standardized tasks, such as visual anomaly detection or utility load management, trained on synthetic or historical data. These models typically require only fine-tuning before deployment. In contrast, heterogeneous and high-risk sectors require customized solutions and significantly more training.

Qualitative findings from multiple studies highlight a fundamental trade-off faced by manufacturers as they try to balance the development of complex and highly accurate AI models with the need to reduce training time and energy consumption. A key challenge in this process is the dependence of deep learning models on large and high-quality datasets. This dependence makes it difficult to incorporate domain expertise, especially when labeled data is scarce, which is common in industrial settings. The limited availability of labeled training data further complicates model development. Labeling industrial datasets often require expert knowledge, making the process both time-consuming and costly. As a result, training datasets tend to be narrow in scope and do not fully capture the variability of real-world manufacturing environments. This is particularly important given the inherent diversity in manufacturing processes, which includes differences in operating conditions, product types, and defect classifications. Capturing such variability is essential to develop AI models that generalize well across a wide range of tasks.

Table 5 Estimated training times for AI models in specific manufacturing use cases.

Use case	AI techniques used	Estimated training duration	Source
AI-driven optimization of steam power boiler combustion using by-product gases in steel manufacturing	Random Forest (RF), Classification and Regression Tree (CART)	~4 months *(authors' assumption[1])*	[91]
AI-assisted process control, kiln stabilization, and optimization of fuel mix and grinding in cement manufacturing	Artificial Neural Networks (ANN)	1–3 months	[70]
AI-driven process optimization for detecting energy use anomalies, load management issues, phase imbalances, and more in building components manufacturing	Recurrent Neural Networks (RNN)	2 weeks	[106]
AI-driven optimization of key operating parameters, including motor load, pulp consistency, and refiner plate gap, to support energy reduction strategies in pulp and paper manufacturing	ANN, Adaptive Neuro-Fuzzy Inference System (ANFIS), Bayesian Optimization	>2–4 weeks *(authors' assumption[2])*	[89]
AI-driven visual defect detection in automotive manufacturing	Convolutional Neural Networks (CNN)	<1 day	[93]

[1] The model was trained on real-time operational data collected between December 2020 and March 2021 [91]. Based on the overlap between data collection and deployment, we assume that model training and refinement may have continued during this period, although the exact schedule is not reported.
[2] The model reportedly undergoes a 2–4-week validation trial on the production line, during which its real-time performance is monitored and systematic errors are corrected through iterative refinement of the model layers [89]. In the absence of a reported development timeline, we assume that the primary period of on-line training and refinement corresponds to this 2–4-week trial window, while recognizing that additional offline development likely occurred outside this interval.

Overfitting represents another critical concern in this context. It occurs when a model performs exceptionally well on training data but fails to generalize to new inputs. This problem is especially common in manufacturing where datasets can be uniform, and tasks are usually narrowly defined. High training accuracy in these settings can be misleading because it often reflects over tuning to the training data rather than true model robustness [19]. Therefore, even small changes in input or context can cause model performance to drop significantly. Finally, the complex and dynamic nature of manufacturing systems leads to changes in setpoints that AI models must be able to handle. These changes require resilient modeling strategies that can adapt to varying conditions and ensure consistent performance in real world applications.

3.2. Case Study: Net Energy Impacts of AI Applications in Cement Manufacturing

The cement industry faces significant challenges, including the need to reduce energy demand and CO_2 emissions while managing complex production processes involving secondary raw materials and alternative fuels. AI offers promising solutions to address these issues. By continuously updating historical data with real-time inputs, AI systems can refine predictive models to support future decision-making.

When integrated with process control systems, these tools can help advance operations toward greater autonomy and improved energy efficiency.

In cement manufacturing, as shown in Table 3, ANNs combined with optimization algorithms can identify optimal process setpoints, including raw material and fuel mixes. AI models iteratively predict ideal operating conditions and recommend fuel mix adjustments to stabilize the kiln and enhance clinker quality. To reduce reliance on fossil fuels and lower operating costs, manufacturers are increasingly adopting alternative fuels such as biomass, solid and liquid waste fuels, waste tires, and plastics. These fuels vary significantly in moisture content, volatility, and particle size, adding further complexity to process control. AI helps manage this complexity by simulating the effects of different fuel blends and learning how they influence overall performance. More specifically, AI improves the monitoring and control of alternative fuel substitution rates, ensuring consistent combustion while minimizing the risk of instability caused by fluctuations in fuel calorific value. This enables informed, data-driven decisions that enhance process efficiency, reduce emissions, and optimize fuel utilization.

For cement grinding, AI-driven models are used to predict and optimize mill performance. Key variables include the material feed rate, separator speed, and main ventilator fan speeds. These predictive models fine-tune parameters that directly impact the quality of the final product. By incorporating AI, cement plants can reduce energy consumption while simultaneously improving product quality and operational efficiency.

The results from these two major AI applications in cement manufacturing are summarized in Table 6. The table provides a quantitative assessment of the balance between energy savings achieved through AI-based process optimization and energy consumption associated with various AI server configurations used for inference, training, and plant control. It presents the net energy savings and corresponding CO_2 reductions for four types of AI servers, ranging from non-accelerated type to those equipped with 2, 4, and 8 GPUs. Each configuration is evaluated under scenarios representing both the lower and upper bounds of reported energy savings in cement plant operations.

The analysis is based on the simplified framework described in Section 2.3, which outlines the net impacts of AI-enabled optimization in manufacturing processes. In a cement plant with a clinker production capacity of 1 Mt per year, estimated gross energy savings may range from 35.83 GWh_{th} to 109.44 GWh_{th} annually, depending on plant configuration, operating conditions, and the application areas. The analysis also accounts for the energy required to run AI models, which vary depending on server specifications and the number of servers deployed. A non-accelerated AI server with a base-case PUE of 1.5 is estimated to consume approximately 0.02 GWh_{th} per year for inferencing, whereas servers equipped with 2, 4, and 8 GPUs require about 0.04, 0.06, and 0.013 GWh_{th} per year, respectively. These figures suggest that AI energy demand increases with more powerful hardware, particularly in high-performance GPU systems.

Despite variations in server energy use, the net energy savings calculated as the difference between gross plant energy savings and AI infrastructure (or MDC) energy demand remain consistently positive. Although the server energy demand for inferencing and periodic training may be overestimated in an industrial context, net gains remain positive even under these conservative assumptions and the lowest reported savings, ranging from 35.65 GWh_{th} per year with an 8-GPU server to 35.80 GWh_{th} per year with a single non-accelerated server. This indicates that the energy required for inference and periodic training is negligible relative to modeled process energy savings under current assumptions. In the upper-bound scenario, net savings are estimated to be even greater, ranging from 109.32 GWh_{th} to 109.42 GWh_{th}

annually across all server types. When also accounting for AI model pre-deployment training energy demand, the first-year net energy savings decreased only marginally, as shown in Table 6.

A sensitivity analysis explores a hypothetical scenario in which five high-powered 8-GPU servers operate simultaneously with a PUE of 2. Even under this extreme assumption, AI-related energy use accounts for only approximately 1–2% of the total gross energy savings. Although the absolute AI energy demand may appear substantial, it is estimated to remain small relative to the overall efficiency gains enabled by process optimization. To explicitly account for the uncertainty, a Monte Carlo simulation is performed for Year 1 (including pre-deployment training energy) and for Year 2 onward (regular inferencing and on-site periodic training), as described in Section 2.3. Key variable inputs, including relative process energy savings, server count, GPU configuration, PUE, and training duration, are represented using probability distributions to capture plausible operating conditions. The Year 2 simulation yielded a mean net energy savings of approximately 60.03 GWh_{th} per year, with a 5^{th} - 95^{th} percentile range of 37.22 - 92.11 GWh_{th} per year. Using the triangular distribution for Year 1, with the most-likely AI energy demand increased from 0.17 to 0.21 GWh per year, the mean net energy savings remained essentially unchanged at approximately 60.01 GWh_{th} per year, with an updated 5^{th} - 95^{th} percentile range of 37.20 - 92.10 GWh_{th} per year. Compared with the Year 2 case, the decrease in mean net energy savings is estimated to be negligible. These results indicate that, even under conservative assumptions regarding process energy savings, the energy demand associated with AI workloads remains small relative to the magnitude of energy savings achieved through process optimization. For additional context, in the low-savings scenario, it would require on the order of approximately 200 continuously operating 8-GPU servers with a PUE of 2 for the associated AI energy demand to offset the gross energy savings achieved through kiln and grinding system optimization. This threshold would increase to approximately 250 servers at a PUE of 1.5. Similar thresholds would vary substantially under high-savings scenarios, different server configurations, and varying levels of deployment.

From another perspective, cement grinding accounts for over half of the reference plant's electricity use i.e. approximately 57 GWh_e per year. A 10% reduction in this demand, achieved through the AI system running on an 8-GPU server that consumes 0.17 GWh_e annually, would result in net electricity savings of approximately 5.53 GWh_e per plant per year. In 2024, the U.S. produced 77 million tonnes of clinker across 92 cement plants [107], with an average production of 0.84 million tonnes per plant. If all U.S. plants adopted this AI application for cement grinding, the total net energy savings would be sufficient to power a large-scale ~50 MW_e data center. It is important to note that this potential reflects energy savings from cement grinding alone while the greatest opportunities lie in optimizing the cement kiln system, where significantly larger amounts of fuel could be saved and made available for other uses.

This study also translates net energy savings into CO_2 abatement, reflecting avoided emissions from reduced combustion-related energy consumption. Across both lower- and upper-bound scenarios, net emission reductions are estimated to range from approximately 3-9%. However, because process emissions are governed by process chemistry and remain unaffected by energy efficiency improvements (see Section 2.3), the abatement potential decreases to approximately 1–4% when expressed relative to total plant CO_2 emissions, including both process and combustion sources, though it remains meaningful. Importantly, even the server configuration with the highest energy demand is still estimated to yield substantial net combustion-related CO_2 reductions, further reinforcing the environmental benefits of AI-driven optimization. Furthermore, given that cement grinding represents only a fraction of the overall primary energy demand (i.e., 12%; Figure 2), with an electricity savings potential of 5–10% for that unit operation, assuming an overall grid mix that includes nuclear and renewable energy sources has only a limited impact on the estimated CO_2 abatement potential. However, these estimates are based on the first-order framework described in Section 2.3 rather than plant-validated assessments, highlighting the need for future work.

Table 6 Estimated net energy and emissions savings in the cement manufacturing process under various assumptions.

	Unit	Estimated plant energy savings		AI-server type			
		Lower bound	*Upper bound*	*8 GPUs*	*4 GPUs*	*2 GPUs*	*Non-accelerated*
Plant Energy Savings (ES_P) [1,2]							
Kiln system	% of unit operation energy demand (ED_U)	3%	10%				
	GWh_{th} per year	28.33	94.44				
Cement grinding	% of unit operation energy demand (ED_U)	5%	10%				
	GWh_{th} per year	7.50	15.00				
AI Infrastructure (or MDC) Energy Demand (ED_{AI}) [1,3]							
PUE 1.5							
1 server	GWh_{th} per year			0.13	0.06	0.04	0.02
5 servers	GWh_{th} per year			0.63	0.29	0.21	0.12
PUE 2							
1 server	GWh_{th} per year			0.17	0.08	0.06	0.03
5 servers	GWh_{th} per year			0.84	0.39	0.28	0.16
Net Energy Savings (ES_N) [1]							
Year 1 (accounting for pre-deployment training energy) [4]							
Lower Bound (MCD PUE = 2; Training period = 3 months)							
1 server	GWh_{th} per year			35.62	35.71	35.74	35.76
5 servers	GWh_{th} per year			34.95	35.40	35.52	35.64
Upper Bound (MCD PUE = 1.5; Training period = 1 month)							
1 server	GWh_{th} per year			109.30	109.37	109.39	109.41
5 servers	GWh_{th} per year			108.80	109.14	109.22	109.31
Year 2 onwards							
Lower Bound (MCD PUE = 2)							
1 server	GWh_{th} per year			35.67	35.76	35.78	35.80
5 servers	GWh_{th} per year			34.99	35.44	35.56	35.68
Upper Bound (MCD PUE = 1.5)							
1 server	GWh_{th} per year			109.32	109.39	109.40	109.42
5 servers	GWh_{th} per year			108.81	109.15	109.24	109.33
Net CO_2 Abatement (CA_N)							
Year 1 (accounting for pre-deployment training emissions) [4]							
Lower Bound (MCD PUE = 2; Training period = 3 months)							

1 server	kt CO_2 per year		11.91	11.94	11.95	11.96
5 servers	kt CO_2 per year		11.70	11.85	11.88	11.92
Upper Bound (MCD PUE = 1.5; Training period = 1 month)						
1 server	kt CO_2 per year		36.76	36.78	36.78	36.79
5 servers	kt CO_2 per year		36.60	36.70	36.73	36.76
Year 2 onwards						
Lower Bound (MCD PUE = 2)						
1 server	kt CO_2 per year		11.93	11.96	11.96	11.97
5 servers	kt CO_2 per year		11.72	11.86	11.89	11.93
Upper Bound (MCD PUE = 1.5)						
1 server	kt CO_2 per year		36.76	36.78	36.79	36.79
5 servers	kt CO_2 per year		36.60	36.71	36.73	36.76

[1] *All energy values are expressed in primary energy terms, using a primary energy factor of 2.6 based on the U.S. EIA 2023 Annual Energy Outlook (AEO).*
[2] *Cement plant capacity is assumed to be 1 Mt of clinker per year with an annual primary energy demand of 1,210 GWh_{th} and corresponding energy-related CO_2 emissions of 0.4 $MtCO_2$.*
[3] *The AI server energy demand is estimated based on sensitivity cases in which 1 to 5 servers (either with 2 to 8 GPUs or non-accelerated) are required for the AI use case in a single plant.*
[4] *An 8-GPU server running continuously for 1–3 months, with a PUE of up to 2, results in 0.01–0.04 GWh of primary energy demand for AI model training and corresponding 4.3–12.9 tCO_2 emissions, which should be subtracted from net energy and CO_2 savings, respectively, in all scenarios for the first year after AI deployment.*

Furthermore, the manufacturing of AI hardware entails embodied CO_2e emissions. For example, NVIDIA reports a cradle-to-gate carbon footprint of approximately 1.3 tCO_2e for its HGX H100 GPU baseboard (as also considered in this study), with materials and components accounting for about 91% of the total [108]. Under a conservative assumption of up to five deployed servers, the associated embodied emissions would reduce the net CO_2 abatement reported in Table 6 by approximately ~0.01 kt CO_2 per refresh cycle for all five servers combined (assuming a 5-year refresh interval).[1] Over a 40-year cement kiln and grinding mill lifetime, this corresponds to seven server refresh cycles (with no refresh in the final five years), resulting in a total of ~0.05 kt CO_2 from repeatedly refreshing the same five servers over the equipment lifetime. While this estimate is subject to uncertainty in hardware configuration, deployment scale, and lifecycle assumptions, it suggests that embodied emissions are likely small relative to the net CO_2 reductions achieved through process optimization. However, it should be noted that life-cycle considerations are outside the scope of the present analysis, and this discussion of embodied carbon is provided only for context. As outlined in Section 2.3, embodied energy and emissions associated with AI hardware, often characterized by relatively short lifetimes and rapid obsolescence cycles of 3–5 years [109,110], are not explicitly accounted for. A more comprehensive evaluation would require a full life-cycle assessment and should be addressed in future work.

The AI energy demand presented in Table 6 reflects the average power requirements for both inference and periodic training. As detailed in Section 3.1.3, the AI model typically requires an initial off-site training phase prior to deployment, which in the context of cement manufacturing is reported to take approximately 1–3 months. For this case study, and in the absence of more precise data, it is conservatively assumed that a high-performance GPU-based server operates continuously during this period, consuming energy comparable to that of the 8-GPU server shown in Figure 3. The training energy demand (0.01–0.04 GWh_{th}) and corresponding CO_2 emissions (4.3–12.9 tCO_2) during this pre-deployment training period must be accounted for in the net energy and CO_2 savings only in the first year of operations after AI deployment, as presented in Table 6. Even with this added pre-deployment training period, the resulting increase in electricity demand is estimated to be relatively minor when considered over the first year of net energy savings. From year 2 onwards, net energy savings are estimated to remain of the same order of magnitude as presented in Table 6 under various scenarios. Hence, the training energy for AI models used in cement process optimization, although significant, is not estimated to substantially affect the overall net energy benefit for cement plants. However, this conclusion is specific to the sector studied and should be reassessed on a process-by-process basis.

4. Overall Discussion and Limitations

The case study results are encouraging and, under the stated assumptions, suggest a net positive effect of AI solutions on the overall cement manufacturing process, beyond what traditional control systems can achieve. For perspective, Zuberi et al. [16] previously evaluated optimized process control based on model predictive control and fuzzy logic in cement production and reported primary energy savings of approximately 3%. In contrast, this study, based on reported savings from advanced AI-enabled approaches (e.g. neural networks and analytical AI), estimates net primary energy reductions of up to 9% for the same dry cement process. This highlights the additional impact that AI, inclusive of improvements that may also stem from rule-based optimization, can bring to process control and overall efficiency. The more recent U.S. DOE Transformative Pathways study [111] estimates overall primary energy savings of

[1] 1.3 tCO_2 per server per refresh cycle × 5 servers (maximum) = 6.5 tCO_2 per refresh cycle (0.0065 $ktCO_2$). This estimate should then be multiplied by the number of server refresh cycles over the plant lifetime to determine the total embodied emissions from AI servers.

approximately 22% relative to Best Available Technology (BAT) cement plants, considering a range of measures including cement blending, preheater/precalciner retrofitting, and other interventions. Although the results of this study are not directly comparable due to potential differences in AI models for retrofitted facilities, achieving up to 9% savings through AI in current typical U.S. cement plants is equivalent to roughly 40% of the energy efficiency improvement needed to reach BAT levels.

These findings suggest that AI-enabled energy optimization in cement manufacturing (and potentially other industrial processes) can deliver significant efficiency and environmental benefits. While assumptions regarding server power, PUE, and training duration can influence the net-benefit range, the relatively low AI energy use suggests that a broad range of server configurations can still yield positive outcomes. Further improvements in AI hardware efficiency could enhance these benefits even more. Nevertheless, while the current research provides a valuable foundation for understanding AI applications in manufacturing, it is subject to several analytical limitations, which are acknowledged and discussed below:

- One of the most significant gaps is the absence of a comprehensive cost analysis associated with AI infrastructure and control system deployments. The initial investment required for enhanced digitalization, such as sensor integration, data management systems, and connectivity upgrades, can be substantial. These upfront costs have not been systematically assessed in this study, yet they represent a critical factor influencing the economic viability and scalability of AI applications. Future research must incorporate robust techno-economic analyses that consider both capital and operational expenditures associated with AI adoption.

- Another major analytical gap lies in the estimation of computational requirements. Due to a lack of detailed information on hardware specifications, energy usage, and data processing needs across diverse AI applications, this study relied on generalized assumptions. These limitations hinder the accuracy of net energy impact assessments and cost estimations. To improve future modeling efforts, a more granular understanding of the computational footprint, including server type, energy usage intensity, and model architecture, must be developed through empirical plant-level data collection and industry collaboration.

- While this work focuses primarily on net energy benefits, based on reported savings rather than direct measurements, which is a limitation in itself, it does not systematically account for the broad range of non-energy impacts that AI deployment may deliver, including embodied carbon effects. These impacts include improved operational productivity, extended equipment lifetimes through predictive maintenance, enhanced workplace safety, better compliance with environmental and safety regulations, and overall improvement in work environments. In several real-world use cases, such non-energy benefits may be the primary drivers for AI adoption, particularly when direct energy savings are marginal. Capturing these multifaceted impacts in future studies is essential for more holistic evaluations of AI's overall value proposition.

- The research is also limited by its sectoral scope. While cement manufacturing was selected as the focal case study due to the relative maturity of AI applications and the presence of a robust ecosystem of AI service providers, the findings on impacts may not readily extrapolate to other heavy manufacturing sectors. In industries such as steel, glass, and chemicals production, AI deployment remains at the early research or pilot stage and is often restricted to narrow, customized applications. The heterogeneity in production processes and data infrastructure across these sectors poses additional challenges. As AI technologies and data availability evolve, further sector-specific case studies will be necessary to assess feasibility, scalability, and net benefits in varied industrial contexts.

- The cement case study at the U.S. sectoral level relies on national averages, such as grid mix, plant fuel mix, and system configuration, to derive weighted-average energy use and emission intensities. Expanding the scope to include geospatial analysis of individual plants across different U.S. states, as well as extending the assessment to other countries with different cement manufacturing technologies and AI adoption opportunities, would further strengthen and broaden the estimated range of net energy impacts.

- The successful implementation of AI also depends on addressing key adoption barriers spanning technical, organizational, and policy dimensions. As detailed in Table 7, overcoming these challenges will require coordinated action from multiple stakeholders. Insights from this study further contextualize these barriers. In particular, the limited availability of transparent and standardized data on AI computational requirements and energy use, as identified in this work (see Section 2.3), directly reinforces challenges related to data quality, availability, and measurement. This lack of data contributes to uncertainty in evaluating net energy and environmental impacts, which may in turn increase perceived operational risks. Furthermore, specialized AI service providers, as highlighted in the qualitative assessment (see Sections 3.1.1-2), suggest how external, domain-focused solutions can help overcome barriers such as limited in-house expertise, high implementation complexity, and difficulties integrating with legacy systems. Several providers offer pre-trained models, deployment support, and scalable platforms that reduce the burden on plant operators and lower adoption barriers.

  Building on these insights, plant operators should prioritize the generation and maintenance of high-quality, structured operational data to enable effective model development and reduce implementation risks. In turn, AI service providers should focus on developing scalable, cost-effective solutions compatible with legacy industrial systems while ensuring robust cybersecurity, as concerns in this area may otherwise drive a shift toward on-site AI solutions (refer to the discussion in Section 2.3) that require a higher degree of workforce training as a prerequisite and could slow adoption. They should also play an active role in workforce development by providing training and support for AI integration. Researchers are encouraged to continue advancing AI methodologies, conducting rigorous cost-benefit and impact assessments, and identifying new industrial use cases. Policymakers should foster a supportive regulatory environment through targeted incentives, data governance frameworks, and initiatives that promote AI adoption for techno-economic and environmental benefits.

Table 7 Barriers and challenges to AI adoption in manufacturing industry (Source: [1,14,112]; partially reaffirmed by findings from this study).

Challenges	**Description**
Data quality and availability	Needs clean, structured data, which is often lacking
Operational risks	Some models may lack the precision required for reliable manufacturing use.
Implementation costs	High upfront costs can limit adoption, especially for smaller manufacturers
Integration with legacy systems	Ensuring compatibility of AI systems with legacy systems may require significant modifications
Limited workforce	Lack of AI and data science expertise hinders adoption without workforce investment
Cybersecurity	Increased connectivity raises the risk of cyberattacks, requiring stronger security measures

In summary, while this study lays important groundwork, addressing the outlined limitations and adoption challenges through targeted future research and stakeholder collaboration will be essential to fully realizing AI's transformative potential in manufacturing. AI deployment should be evaluated from a net, system-level perspective rather than by considering its computational energy burden in isolation. When applied to real-time operations, AI can serve as an effective strategy for improving energy efficiency and reducing emissions, while also delivering non-energy benefits such as productivity gains, enhanced system reliability, extended equipment lifetimes, and improved operational and quality control. As such, the qualitative and quantitative findings of this study have important implications for both policy and practice.

Policymakers should adopt performance-based, net-impact assessment frameworks and support AI development, demonstration, and measurement. Such a framework for AI in manufacturing could build on existing industrial energy management and reporting standards by integrating AI-specific energy and emissions accounting. For example, ISO 50001 provides a structured approach for continuous monitoring and verification of energy performance, while cap-and-trade systems establish robust protocols for emissions reporting and verification. Incorporating AI infrastructure monitoring into an ISO 50001 energy review could involve defining AI systems (e.g., kiln optimization models or predictive maintenance algorithms) as significant energy uses (SEUs), installing sub-metering for associated compute infrastructure (on-site servers or cloud-equivalent allocations), and integrating AI-related energy consumption and performance improvements into the facility's energy baseline and energy performance indicators (EnPIs). For instance, a cement plant deploying an AI-based kiln control system would track the additional electricity consumed by model training and real-time inference alongside reductions in kiln fuel intensity (e.g., GJ/t clinker), enabling calculation of net energy and emissions impacts within the ISO 50001 continuous improvement cycle. More broadly, extending these frameworks to AI applications would require: (i) tracking incremental electricity use from AI infrastructure (training and inference); (ii) quantifying process-level energy and emissions savings attributable to AI deployment; and (iii) reporting net energy and CO_2 impacts on a consistent, per-unit-output basis, thereby enabling transparent, comparable, empirical, and verifiable assessments of AI's net impact in industrial settings.

Manufacturers, in turn, should view AI not merely as a digital upgrade but as a complementary strategy to conventional efficiency measures for both existing and new facilities. Although further plant-validated evidence is needed, the results indicate that well-designed AI deployments, particularly those using on-site or hybrid architectures, can deliver robust net benefits and should be considered as part of industrial energy strategies.

5. Conclusions

In conclusion, while the rising energy consumption of AI systems raises valid concerns, this study suggests that AI deployment in manufacturing, particularly through deep learning and predictive analytics, has the potential to deliver net energy and environmental benefits when evaluated at the system level. The analysis suggests that AI can improve process efficiency, reduce operational costs, and enhance productivity beyond the capabilities of traditional control systems, although these effects depend on deployment context and operational conditions. The cement manufacturing case study indicates potential reductions in net primary energy use and associated CO_2 emissions of approximately 3-9%. However, these estimates are derived from a first-order analytical framework rather than plant-validated assessments and should therefore be interpreted as indicative rather than definitive, underscoring the need for rigorous empirical validation in real-world industrial settings.

The proposed first-order framework is generally applicable to other energy-intensive manufacturing sectors, provided that sector-specific inputs, such as baseline energy demand, process-level efficiency improvements, AI deployment configurations, and associated computational requirements, are available. While the current study demonstrates the approach using the cement sector, its broader application is primarily constrained by data availability and quality rather than by methodological limitations.

Several limitations of this study warrant consideration. A comprehensive techno-economic assessment was beyond the scope of this work, and capital and operational costs associated with digital infrastructure were not evaluated. The analysis also relies on generalized assumptions regarding AI computational requirements due to limited plant-level data, which introduces uncertainty in the estimated net energy savings. In addition, the focus on energy impacts does not capture broader non-energy benefits.

With these constraints, the relatively modest energy demand of AI inference and periodic training, compared to the magnitude of process efficiency improvements estimated in the cement case study, suggests that similar net benefits may be achievable in other energy-intensive manufacturing processes, subject to validation. Overall, these findings highlight the importance of evaluating AI deployment from a net, system-level perspective and provide a structured basis for further empirical and techno-economic analysis, with insights that may inform industry stakeholders and policymakers in guiding effective and responsible AI adoption in manufacturing.

CRediT authorship contribution statement

M. Jibran S. Zuberi: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft. **Haitam Laarabi:** Conceptualization, Visualization, Writing – review & editing. **Sarah Smith:** Conceptualization, Data curation, Writing – review & editing. **Arman Shehabi:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing – review & editing.

Acknowledgments

The work described in this study was conducted at Lawrence Berkeley National Laboratory and funded by Google under Project ID 110439. The authors would like to thank Jon Turnbull and Antonia Gawel from Google for their discussions and helpful advice during this work. The views and opinions expressed herein are those of the authors and do not necessarily reflect those of Google, the United States Government or any agency thereof, or The Regents of the University of California.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 5.2 and Grammarly to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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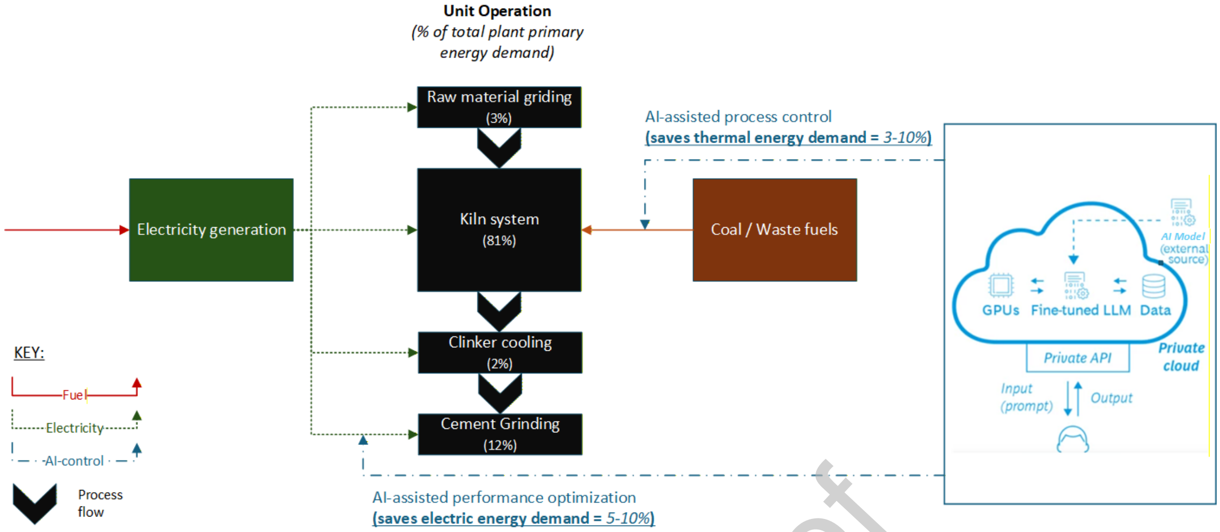
Graphical abstract



Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: